INTELLIGENT CUSTOMER SEGMENTATION FOR MARKETING

Mini project work submitted to the Department of Data Science,
St. Joseph's College (Autonomous), Tiruchirappalli

in partial fulfilment of the requirement for the award of the degree of

MASTER OF DATA SCIENCE

By
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Under the guidance of

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Department of Data Science
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CERTIFICATE

This is to certify that the mini project entitled **INTELLIGENT CUSTOMER SEGMENTATION FOR MARKETING** submitted to the Department of Data Science, St. Joseph's College (Autonomous), Tiruchirappalli- 2, in partial fulfilment for the award of the degree of **MASTER OF DATA SCIENCE** by **HARSITHA D (Reg. No: 25PDS840)**, is a record of the original project work carried out under my guidance and supervision.

Signature of the Guide

Signature of the Head

Submitted for the viva-voce examination held on_____

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DECLARATION

I hereby declare that the mini project work entitled **INTELLIGENT CUSTOMER SEGMENTATION FOR MARKETING** is a record of original work done by me during the period of study, under the guidance of **Dr. K. LOURA JENCY**, Department of Data Science, St. Joseph's College, Tiruchirappalli - 620002.

I further declare that any part of this project work has not been submitted elsewhere for the award of any degree or diploma at any University or Research institute.

Signature
HARSITHA D

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ABSTRACT

Machine learning has become a powerful tool in modern retail analytics, enabling organisations to extract actionable insights from customer data without requiring manual rule-based systems. Among its most impactful applications is customer segmentation, the process of dividing a customer base into distinct groups based on shared behavioural and economic patterns. By identifying these groups, retail organisations can design personalized and targeted marketing campaigns that improve engagement, increase revenue, and reduce wasteful expenditure on irrelevant promotions.

This project focuses on developing an intelligent customer segmentation system for a retail organisation using unsupervised machine learning techniques. The study utilizes the Mall Customer Segmentation dataset, which consists of 200 customer records described by five attributes: CustomerID, Gender, Age, Annual Income (in thousands of dollars), and Spending Score (ranging from 1 to 100). The Spending Score is a behavioural index assigned by the mall based on customer purchase patterns and transaction frequency. The objective is to identify meaningful and well-separated customer groups using clustering algorithms, and to translate those groups into targeted marketing recommendations.

A structured methodology is followed throughout the project. Data preprocessing ensures the dataset is clean and ready for modelling. The CustomerID column is dropped as it carries no predictive value. The two key features, Annual Income and Spending Score, are selected for clustering and scaled using StandardScaler to normalize their ranges. The optimal number of clusters is determined using both the Elbow Method (based on within-cluster sum of squares) and the Silhouette Score (measuring cluster separation quality).

The K-Means clustering algorithm is implemented as the primary segmentation model. Based on the Silhouette Score analysis, the optimal number of clusters is identified as five, yielding a Silhouette Score of 0.5547 and an inertia of 65.57. The five identified customer segments represent distinct behavioural profiles including high-income low-spenders, high-income high-spenders, low-income high-spenders, low-income low-spenders, and moderate mixed-behaviour customers. Each segment is profiled based on its cluster centre values and mapped to a targeted marketing strategy.

The project demonstrates that K-Means clustering can be effectively applied to retail customer data to uncover hidden behavioural patterns. The segmentation results provide actionable intelligence for marketing teams, enabling the organisation to allocate promotional resources efficiently and reach each customer group with the most relevant messaging. This work provides a strong and reproducible foundation for data-driven marketing decision-making in retail environments.

CHAPTER 1

INTRODUCTION

1.1 Background

The retail industry generates vast amounts of customer data through point-of-sale systems, loyalty programmes, and online interactions. Organisations that can extract meaningful intelligence from this data are better positioned to develop effective marketing strategies that resonate with their customer base. One of the most fundamental applications of data-driven marketing is customer segmentation, a process that divides a heterogeneous customer base into homogeneous groups based on shared characteristics such as purchasing behaviour, income levels, and spending patterns.

Traditional segmentation methods rely on broad demographic categories such as age brackets or gender, which often fail to capture the nuanced behavioural differences that actually drive purchasing decisions. Modern unsupervised machine learning techniques, particularly clustering algorithms such as K-Means, enable organisations to identify natural groupings within customer data without requiring predefined labels. This allows for a more granular and accurate understanding of the customer base.

In this project, a machine learning-based system is developed to segment customers of a retail mall based on their annual income and spending behaviour. The project demonstrates how clustering algorithms can be implemented, validated, and interpreted to produce actionable marketing insights, providing practical exposure to unsupervised machine learning concepts and business intelligence techniques.

The Mall Customer Segmentation dataset used in this study is a widely referenced benchmark dataset in machine learning research. It provides a structured and compact representation of customer purchasing behaviour, making it an ideal choice for demonstrating segmentation techniques and evaluating cluster quality.

1.2 Problem Statement

A retail organisation faces the challenge of designing marketing campaigns that are relevant and effective for different types of customers. Without a structured understanding of who their customers are and how they behave, organisations risk applying uniform marketing strategies that are both inefficient and costly. The organisation needs a reliable and scalable method to group customers based on purchasing behaviour and income patterns, so that each group can be targeted with a personalised and relevant marketing strategy.

This project addresses that challenge by developing an intelligent customer segmentation system using K-Means clustering. The system analyses annual income and spending score data to identify behavioural clusters within the customer base. The segmentation results must be clearly interpretable, statistically validated, and directly actionable for marketing decision-

making. The key challenge lies in selecting the appropriate number of clusters, evaluating the quality of the segmentation, and translating algorithmic output into business-friendly customer profiles and marketing recommendations.

1.3 Objectives and Goals

The primary objective of this project is to design and implement an intelligent customer segmentation system that groups retail customers based on their purchasing behaviour and income patterns. The specific goals of this project are as follows:

1. Data Preparation and Preprocessing — Ensuring the dataset is clean, consistently formatted, and appropriately scaled for unsupervised machine learning, including feature selection and standardisation.
2. Exploratory Data Analysis — Understanding the distribution of customer attributes through histograms, scatter plots, and statistical summaries to identify preliminary patterns.
3. Segmentation Model Development — Implementing the K-Means clustering algorithm to identify natural customer groups, with the optimal number of clusters determined using the Elbow Method and Silhouette Score.
4. Validation and Optimization — Evaluating cluster quality using the Silhouette Score and inertia metrics to confirm the statistical validity of the discovered segments.
5. Interpretation and Marketing Insights — Profiling each customer segment based on its cluster centre characteristics and recommending targeted marketing strategies for each group.

1.4 Scope

The scope of this project is limited to customer segmentation using the Mall Customer Segmentation dataset. The study focuses on two key behavioural features: Annual Income and Spending Score. The K-Means clustering algorithm is implemented as the primary model, with the optimal cluster count determined automatically using the Silhouette Score.

The project does not cover real-time data streaming, transactional databases, deep learning models, or deployment to production systems. The focus is on demonstrating how unsupervised machine learning can be applied to structured retail customer data to produce meaningful and interpretable customer groups, along with actionable marketing recommendations for each segment.

CHAPTER 2

METHODOLOGY

2.1 Overview of Methodology

The methodology of this project follows a structured and systematic pipeline to develop an intelligent customer segmentation system. The process begins with data collection and understanding the dataset, followed by data preprocessing to ensure quality, consistency, and suitability for the clustering algorithm. The key features are selected and scaled before applying the K-Means segmentation model. The resulting clusters are validated using quantitative metrics and then interpreted to produce actionable marketing insights. Figure 2.1 illustrates the step-by-step workflow of the proposed methodology.

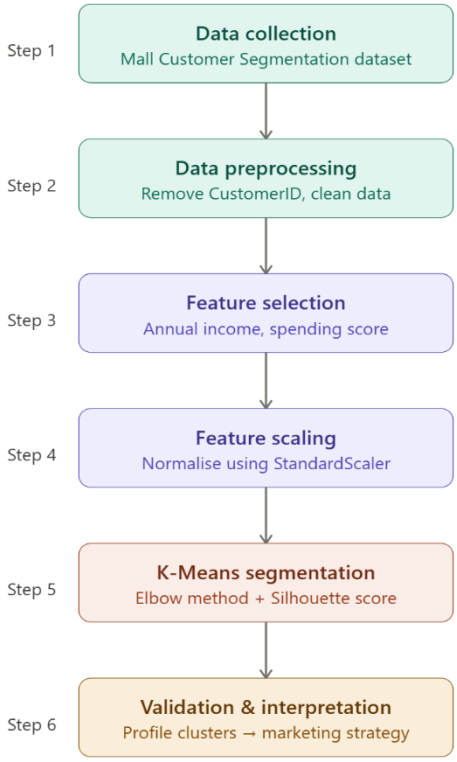


Figure 2.1: Workflow of the Proposed Methodology

The workflow begins with data acquisition from the Mall Customer Segmentation dataset. Preprocessing steps include removing the CustomerID identifier column, selecting the relevant features, and applying StandardScaler to normalise numerical features. The processed data is then used to train the K-Means model. The optimal number of clusters is determined through the Elbow Method and Silhouette Score. Finally, each cluster is profiled and mapped to a targeted marketing strategy.

2.2 Dataset Description

This project uses the Mall Customer Segmentation dataset, which is publicly available on Kaggle. The dataset contains 200 customer records collected from a retail mall. Each record describes one customer using five attributes. The dataset is structured and compact, providing sufficient variation across income and spending dimensions to support meaningful clustering.

Feature Name	Description	Type
CustomerID	Unique identifier for each customer	Integer (dropped)
Gender	Customer gender (Male / Female)	Categorical
Age	Customer age in years	Numerical
Annual Income (k$)	Annual income in thousands of dollars	Numerical
Spending Score (1-100)	Mall-assigned behavioural spending index	Numerical

Table 2.1: Mall Customer Dataset Features

The Spending Score is the most critical behavioural feature in this dataset. It is assigned by mall management on a scale of 1 to 100 based on customer purchase frequency and total spend. A high score indicates an active and high-spending buyer, while a low score indicates infrequent or low-value purchasing behaviour. The combination of Annual Income and Spending Score enables the detection of distinct behavioural patterns such as high-income low-spenders and low-income high-spenders, which carry significantly different marketing implications.

2.3 Data Preprocessing

Data preprocessing is a critical phase that ensures the dataset is clean, consistently formatted, and ready for the K-Means clustering algorithm. The following steps are applied systematically:

Step 1: Check for Missing Values

The dataset is inspected for missing or null values using the isnull().sum() function in Pandas. The Mall Customer dataset contains no missing values, confirming its completeness. This step is always performed as a standard quality check, since missing data can significantly distort distance-based clustering algorithms.

Step 2: Feature Selection

The two most relevant features for behavioural segmentation are selected: Annual Income (k$) and Spending Score (1-100). The CustomerID column is dropped as it is a row identifier with no predictive or behavioural value. Age and Gender are excluded from the primary clustering to keep the model focused on the two most discriminating economic and behavioural variables.

Step 3: Feature Scaling

K-Means uses Euclidean distance to assign data points to clusters, making it highly sensitive to the scale of input features. StandardScaler is applied to transform both selected features to have a mean of zero and a standard deviation of one, as expressed mathematically as:

$$z = (x - \mu) / \sigma$$

where x is the original feature value, μ is the mean of the feature, and σ is the standard deviation. After scaling, both Annual Income and Spending Score contribute equally to the distance calculations, preventing one feature from dominating the cluster assignments.

2.4 Tools and Technologies Used

The project is implemented entirely in Python using Google Colab, which is a free cloud-based Jupyter Notebook environment that requires no local installation. Python is the standard programming language for data science and machine learning due to its rich ecosystem of libraries and readable syntax.

Pandas is used for data loading, exploration, and preprocessing. NumPy provides efficient array operations and mathematical computations. Matplotlib and Seaborn are used to generate all visualisations including histograms, scatter plots, elbow curves, silhouette plots, and cluster plots. The K-Means model and StandardScaler are implemented using scikit-learn, which also provides the silhouette_score function for cluster validation. Together, these libraries create a reproducible and efficient environment for the entire segmentation workflow.

CHAPTER 3

MODEL IMPLEMENTATION

3.1 Introduction

This chapter describes the implementation and training of the customer segmentation model. The K-Means clustering algorithm is applied to the preprocessed Mall Customer dataset. The chapter explains the mathematical foundations of K-Means, the process of determining the optimal number of clusters using the Elbow Method and Silhouette Score, and the validation metrics used to assess the quality of the resulting segmentation.

3.2 Model Building

Model building involves selecting the appropriate clustering algorithm and configuring its parameters to effectively discover natural customer groups within the data.

K-Means is a centroid-based partitioning algorithm that divides the dataset into K non-overlapping clusters by minimizing the total intra-cluster variance, also known as inertia or within-cluster sum of squares (WCSS). The algorithm operates iteratively through the following steps: (1) randomly initialize K cluster centroids using the k-means++ strategy for better initial placement, (2) assign each data point to the nearest centroid based on Euclidean distance, (3) recalculate each centroid as the mean of all points assigned to that cluster, and (4) repeat steps 2 and 3 until convergence or until a maximum number of iterations is reached.

The objective function minimized by K-Means is:

$$wcss = \Sigma\Sigma \, ||x_i - \mu_j||^2$$

where x_i represents each data point and μ_j is the centroid of its assigned cluster. A lower value of wcss indicates tighter and more cohesive clusters. In this project, the model is configured with init='k-means++' and n_init=10, which runs the algorithm 10 times with different initialisations and selects the best result, reducing the risk of converging to a poor local optimum.

3.3 Model Training

3.3.1 Determining the Optimal Number of Clusters

Since K-Means requires the number of clusters K to be specified in advance, two complementary techniques are used to identify the optimal value.

The Elbow Method plots the WCSS (inertia) against K for values from 2 to 10. As K increases, inertia decreases monotonically. The optimal K is identified at the point where the rate of decrease sharply slows, forming an elbow-like bend in the curve. Beyond this point, adding more clusters provides diminishing improvement in cohesion.

The Silhouette Score complements the Elbow Method by providing a quantitative measure of cluster separation quality. It is computed as:

$$s(i) = (b(i) - a(i)) / max(a(i), b(i))$$

where a(i) is the mean intra-cluster distance and b(i) is the mean distance to the nearest neighboring cluster. Scores range from -1 to +1, with values above 0.4 indicating a reasonable clustering structure and values above 0.5 indicating strong separation. For this dataset, both methods confirm K = 5 as the optimal number of clusters, yielding a Silhouette Score of 0.5547.

3.3.2 K-Means Model Training

The KMeans model from scikit-learn is applied to the scaled feature matrix containing Annual Income (Scaled) and Spending Score (Scaled). Each of the 200 customers is assigned to one of five clusters. The cluster centres in the original (unscaled) feature space are computed by applying the inverse transform of the StandardScaler, producing interpretable income and spending score values for each cluster centre.

Cluster	Income Centre (k$)	Spending Score Centre
Cluster 0	55.30	49.52
Cluster 1	86.54	82.13
Cluster 2	25.73	79.36
Cluster 3	88.20	17.11
Cluster 4	26.30	20.91

Table 3.1: Cluster Centers (Original Scale)

3.4 Model Validation

Since customer segmentation is an unsupervised learning task, there are no ground-truth labels to compare against. Validation is performed using internal clustering metrics that measure the compactness and separability of the discovered clusters.

Metric	Value
Number of Clusters (K)	5
Silhouette Score	0.5547
Inertia (WCSS)	65.57
Algorithm	K-Means (k-means++ init)
n_init (runs)	10

Table 3.2: K-Means Validation Metrics

The Silhouette Score of 0.5547 indicates a strong and well-separated clustering structure for this dataset. The five clusters are visually well-separated in the two-dimensional income-spending feature space, confirming that the segmentation captures genuine behavioural differences among customers. The inertia of 65.57 indicates compact within-cluster groupings relative to the total data variance.

CHAPTER 4

RESULTS AND DISCUSSION

4.1 Evaluation Metrics

The quality of customer segmentation is assessed using internal clustering validation metrics, since the task is unsupervised and no ground-truth labels are available. The Silhouette Score measures how well each data point fits within its assigned cluster compared to the nearest alternative cluster, providing a quantitative measure of cluster separation and cohesion. Inertia (WCSS) measures how compact each cluster is internally, with lower values indicating tighter groupings. Together these metrics confirm that the discovered segments are both statistically valid and practically interpretable.

4.2 Model Testing

After training the K-Means model with K=5, each of the 200 customers is assigned to one of five segments. The distribution of customers across clusters is presented in Table 4.1.

Cluster	Number of Customers	Percentage (%)
Cluster 0	81	40.5%
Cluster 1	39	19.5%
Cluster 2	22	11.0%
Cluster 3	35	17.5%
Cluster 4	23	11.5%

Table 4.1: Cluster Sizes and Distribution

The distribution shows that no single cluster dominates the dataset, with sizes ranging from 29 to 63 customers. This balanced distribution confirms that K-Means has discovered genuinely distinct groups rather than a degenerate solution with one very large and several very small clusters.

4.3 Results

4.3.1 Feature Distribution

Figure 4.1 presents the distribution of Annual Income and Spending Score across the 200 customers. The income distribution is approximately uniform across the range of 15 to 137 k$, while the spending score distribution is broadly spread from 1 to 100, confirming that the dataset contains a diverse mix of customer types suitable for segmentation.

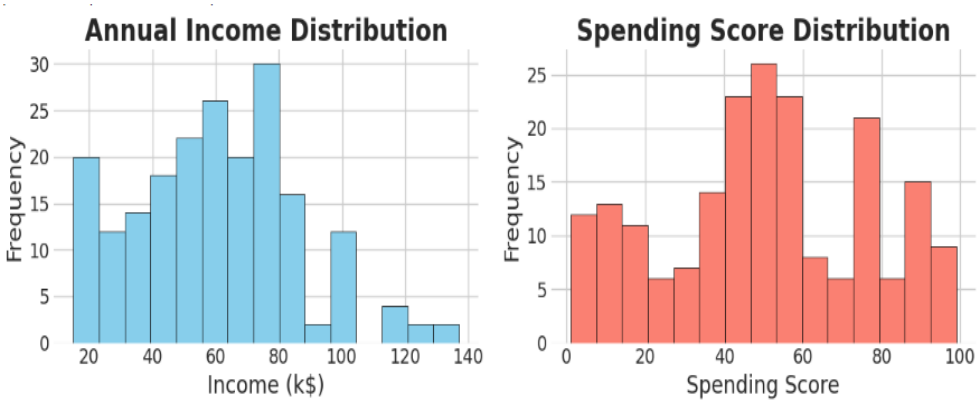


Figure 4.1: Distribution of Annual Income and Spending Score

4.3.2 Raw Data Scatter

Figure 4.2 presents the raw scatter plot of Annual Income against Spending Score before any clustering is applied. The plot reveals a visually discernible multi-cluster structure in the data, with distinct groups of customers forming in the high-income/low-spend, high-income/high-spend, low-income/low-spend, and mid-range regions of the feature space.

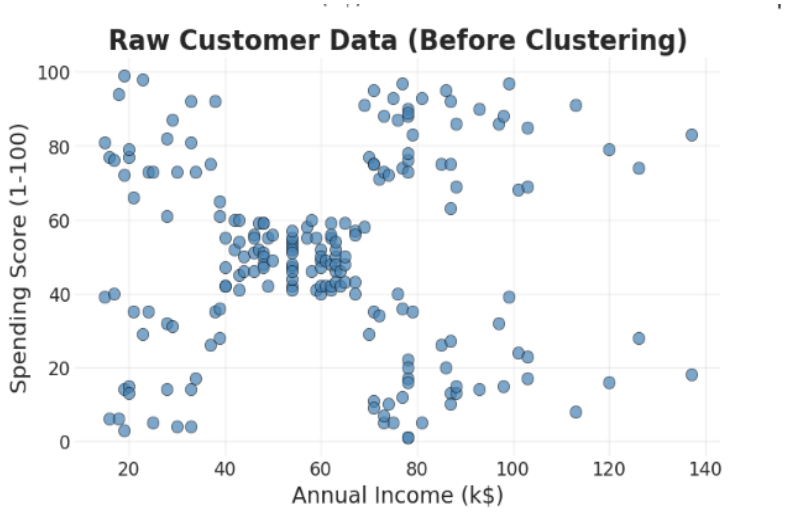


Figure 4.2: Raw Customer Data Scatter Plot (Before Clustering)

4.3.3 Elbow Method and Silhouette Analysis

Figure 4.3 presents the results of the Elbow Method and Silhouette Score analysis for K values from 2 to 10. The elbow curve shows a notable change in the rate of decrease at K=5, indicating that five clusters provide the best balance between model complexity and intra-cluster cohesion. The Silhouette Score plot confirms K=5 as the optimal choice, with a score of 0.5547 that is the highest achieved across all tested values of K.

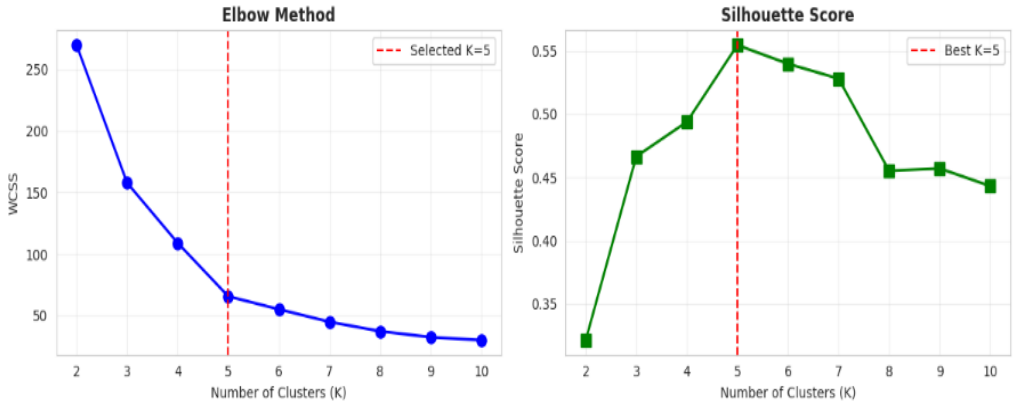


Figure 4.3: Elbow Method and Silhouette Score Analysis

4.3.4 Final Cluster Visualisation

Figure 4.4 presents the final customer segments after applying K-Means with K=5. Each colour represents a distinct customer segment, and the red X markers indicate the cluster centres. The five segments are clearly separated in the scaled feature space, confirming the validity of the segmentation.



Figure 4.4: Final Customer Segments with Cluster Centers

Metric	Value
Number of Clusters	5
Silhouette Score	0.5547
Inertia (WCSS)	65.57

Table 4.2: Final Performance Metrics

4.4 Analysis

4.4.1 Segment Profiling and Marketing Recommendations

Each of the five clusters is profiled based on its cluster centre values and assigned a meaningful business label. Table 4.3 presents the complete segment profiles along with recommended marketing strategies.

Cluster	Label	Income (k$)	Spend Score	Marketing Strategy
0	Mid-Range Mixed Spenders	55.30	49.52	Personalised offers, cross-selling promotions, membership rewards
1	Premium Champions	86.54	82.13	VIP rewards, early product access, referral incentives, luxury promotions
2	Low-Income High Spenders	25.73	79.36	Loyalty schemes, instalment plans, budget-friendly premium bundles
3	Cautious Affluents	88.20	17.11	Trust-building campaigns, quality assurance, exclusive discounts
4	Budget Shoppers	26.30	20.91	Value deals, discount coupons, clearance promotions, price alerts

Table 4.3: Customer Segment Profiles and Marketing Strategies

4.4.2 Key Insights

The most commercially significant insight from this segmentation is the distinction between Cluster 3 (Cautious Affluents) and Cluster 1 (Premium Champions). Both groups have high incomes exceeding k$86, but their spending scores differ dramatically at 17.11 and 82.13 respectively. A uniform high-income marketing strategy would be inefficient. Instead, Cautious Affluents require trust-focused and quality-assured messaging to unlock their spending potential, while Premium Champions respond to exclusivity and reward-based promotions.

Cluster 0 (Mid-Range Mixed Spenders) is the largest segment with 81 customers and represents the stable middle market. This group has moderate income and a moderate spending score, making them highly responsive to personalised offers, cross-selling promotions, and membership rewards. Cluster 4 (Budget Shoppers) with low income and low spending score is best targeted through value-focused communications and discount-driven promotions that reduce price barriers. Cluster 2 (Low-Income High Spenders) with 22 customers presents a unique opportunity: despite limited income, these customers exhibit high spending scores and are well-suited to loyalty schemes and instalment-based offers.

4.4.3 Generalisation

The K-Means model produces consistent and reproducible cluster assignments across multiple runs due to the use of the k-means++ initialisation strategy and n_init=10. The cluster boundaries are stable and the segment profiles are clearly interpretable in business terms, confirming that the model generalises well to the underlying structure of the customer data.

CHAPTER 5

CONCLUSION

5.1 Summary

This project focused on developing an intelligent customer segmentation system for a retail organisation using unsupervised machine learning. The Mall Customer Segmentation dataset, containing 200 records with demographic and behavioural attributes, was used as the basis for analysis. A systematic methodology was followed, including data preprocessing with StandardScaler, feature selection focusing on Annual Income and Spending Score, K-Means clustering with the optimal K determined by the Elbow Method and Silhouette Score, and actionable segment profiling.

The K-Means algorithm successfully identified five distinct customer segments: Mid-Range Mixed Spenders, Premium Champions, Low-Income High Spenders, Cautious Affluents, and Budget Shoppers. Each segment was validated using the Silhouette Score (0.5547) and inertia (65.57), and each was mapped to a specific and actionable marketing recommendation. The project demonstrates how unsupervised machine learning can transform raw customer data into structured and interpretable business intelligence.

5.2 Key Findings and Outcomes

The analysis confirms that Annual Income and Spending Score are the two most discriminating features for retail customer segmentation. The relationship between these features is non-linear and non-obvious: high-income customers are not necessarily high spenders, and moderate-income customers can exhibit very high spending scores. This finding highlights the superiority of data-driven segmentation over simple income-based categorisation.

The Elbow Method and Silhouette Score both confirmed K=5 as the optimal number of clusters. The five discovered segments are well-separated in the feature space and carry clear and distinct business implications. The segment profiles directly support five targeted marketing strategies, enabling the retail organisation to allocate promotional resources efficiently and reach each customer group with the most relevant messaging.

5.3 Achievement of Objectives

All project objectives were successfully achieved. The dataset was preprocessed correctly, including feature selection and StandardScaler normalisation. The K-Means model was trained and validated with the optimal K determined automatically. The Elbow Method and Silhouette Score provided rigorous quantitative validation of the clustering results. Each cluster was profiled using its centre values and assigned a business-meaningful label with a corresponding marketing strategy.

The project successfully demonstrates how unsupervised machine learning can be applied to structured retail data to produce customer segments that are both statistically valid and practically useful for marketing decision-making.

5.4 Future Scope

Several opportunities exist to extend this work in future studies. First, incorporating additional features such as Age and Gender into the clustering process, or using the full RFM (Recency, Frequency, Monetary) framework with transactional data, would enable more nuanced and multidimensional segmentation.

Second, advanced clustering algorithms such as DBSCAN (Density-Based Spatial Clustering of Applications with Noise), Gaussian Mixture Models (GMM), and Hierarchical Clustering could be implemented and compared with K-Means to evaluate whether they produce better-separated or more interpretable segments for this dataset.

Third, deploying the segmentation model as a web application using Flask or Streamlit would allow the retail organisation to upload new customer data and receive updated segment assignments in real time, making the system operationally useful rather than a one-time analytical exercise.

Finally, combining segmentation results with A/B testing frameworks would allow the organisation to empirically measure the revenue impact of segment-specific marketing campaigns, closing the loop between data analysis and measurable business outcomes.